

Scalability Analysis of a Real-Time Hospital Queue Management System using MERN Stack Architecture

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Abstract—In high-density healthcare environments, such as public hospitals in India, traditional manual queuing systems often lead to overcrowding and administrative inefficiencies. This paper presents the design and performance evaluation of a scalable, web-based Hospital Queue Management System developed using the MERN stack (MongoDB, Express.js, React, Node.js). Unlike conventional token systems, our solution offers real-time slot booking and dynamic queue updates. We evaluated the system’s performance using Apache JMeter, simulating concurrent user loads ranging from 100 to 500 users. Experimental results demonstrate a baseline latency of 18 ms at 100 concurrent users and sustained stability at 500 users with 0% packet loss, validating the non-blocking I/O architecture of Node.js as a viable solution for high-throughput medical applications.

Index Terms—Hospital Management System, MERN Stack, Queue Management, Apache JMeter, Scalability, Web Performance.

I. INTRODUCTION

Public healthcare facilities in developing nations face a critical challenge: managing patient flow. Outpatient Departments (OPD) often witness daily footfalls exceeding 500-1,000 patients, leading to chaotic manual queues, lost tokens, and excessively long waiting times [4].

The primary objective of this research is to replace physical queues with a digital “Virtual Queue” system. We propose a web-based application that allows patients to book tokens remotely and track their status in real-time. The core research question addressed in this study is: *Can a standard JavaScript-based stack (MERN) handle the burst traffic typical of hospital morning hours?*

Existing solutions often rely on expensive enterprise software or static websites that fail under load [5]. This paper validates a cost-effective, scalable architecture using Node.js and MongoDB to handle high-concurrency booking requests.

II. SYSTEM ARCHITECTURE

The proposed system is built on a loosely coupled architecture designed for scalability. The application follows the standard Model-View-Controller (MVC) pattern, adapted for the Next.js framework.

A. Frontend

The client-side interface is built with **Next.js (React)**. It provides a responsive mobile-first interface for patients to book appointments. We utilized Client-Side Rendering (CSR) for dynamic queue updates to reduce server load during page navigation.

B. Backend

The API is powered by **Next.js App Router**, running in a Node.js environment. To address the challenge of high concurrency, we implemented a Singleton pattern for the database connection (`dbConnect`). This ensures that the application maintains a connection pool rather than opening a new connection for every request, preventing connection leaks during traffic spikes.

C. Database

MongoDB was chosen for its flexible schema design. The patient model captures essential details including Name, Age, Department, and a calculated Priority level.



Fig. 1. System Architecture Diagram demonstrating the flow from Client to Next.js API to MongoDB.

III. EXPERIMENTAL SETUP

To validate the system’s reliability and scalability, we conducted stress testing using **Apache JMeter (v5.6.3)**. The test environment was configured to simulate an “OPD Rush Hour” scenario on a local server.

A. Test Configuration

- **Test Tool:** Apache JMeter
- **Protocol:** HTTP POST Requests to the API endpoint `/api/patients`.
- **Data Payload:** JSON objects containing patient demographics and department selection.

B. Workload Models

We defined two distinct scenarios to benchmark performance:

- **Scenario A (Baseline):** 100 concurrent users with a 10-second ramp-up period.
- **Scenario B (Stress Test):** 500 concurrent users with a 10-second ramp-up period.

The primary metrics recorded were Average Response Time (Latency), Throughput (Requests/Second), and Error Rate.

IV. RESULTS AND DISCUSSION

A. Performance Analysis

The system performance was evaluated under the two load scenarios described above. Table I summarizes the key metrics obtained from the stress tests.

TABLE I
COMPARATIVE PERFORMANCE METRICS

Metric	Scenario A (100 Users)	Scenario B (500 Users)
Average Latency	18 ms	16 ms
Throughput	10.1 req/sec	50.1 req/sec
Error Rate	0.00%	0.00%
Max Response Time	236 ms	119 ms

B. Scalability Interpretation

The experimental data reveals three critical insights regarding the MERN stack's capability in this context:

- 1) **Throughput Linearity:** As seen in Table I, the throughput increased by exactly $5\times$ (from 10.1 to 50.1 req/sec) when the user load increased by $5\times$. This linear scaling indicates that the Node.js event loop effectively managed the concurrency without bottlenecks or thread blocking.
- 2) **Stability:** The error rate remained at **0.00%** throughout the tests. This confirms that the Mongoose schema validation and API error handling logic functioned correctly even under stress.
- 3) **Latency Efficiency:** The average latency remained under 20 ms in both scenarios. The slight improvement in Scenario B (16 ms) can be attributed to the "warm-up" effect of the MongoDB connection pool, where initial TCP connections were already established from the previous test cycle.

V. CONCLUSION

This study successfully demonstrated the viability of a MERN-stack-based Hospital Queue Management System for handling high-concurrency scenarios. The system maintained data integrity with a **0% error rate** even when the user load increased five-fold. While latency naturally fluctuates under load, the Node.js event-loop architecture prevented server crashes, proving it to be a cost-effective solution for digitizing patient queues.

Future work will focus on implementing horizontal scaling using Docker containers and Kubernetes to distribute load across multiple servers for city-wide deployment.



Fig. 2. JMeter Throughput and Latency Analysis under 500-User Load.

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